Designing and Checking a Spur Gear Pair

A plain language report on building a gear from a text prompt and proving the result with simulation

Prepared by Bertwin Kurisinkal Shine | 7 October 2026

What this report covers

This report is part of a small project where I built a helper that lets me describe a part in plain words and have it modelled in FreeCAD. The helper is an MCP, which is just a bridge that lets the AI talk to the CAD program and run commands inside it. I am not going to spend long on how the bridge works. The short version is that I wanted to check how good it really is, so I asked it to design a real machine part, and then I checked the answer myself using proper engineering simulation.

The part for this test is a spur gear pair. A small electric motor drives the smaller gear, and the pair slows the speed down by three times. This is a normal power transmission job, the kind you would find in a conveyor drive. The aim was to see whether the helper could size the gears sensibly, build them, and whether its strength numbers would hold up when checked independently.

The result in one line: the helper asked the right questions, built a correct and editable gear pair, and when I ran my own finite element analysis in ANSYS the tooth stress it predicted was confirmed, with a large safety margin.

Step 1. The request I gave

I did not hand it a full specification. I gave it the kind of brief an engineer would get from a customer, and left some of the detail open on purpose to see how it would respond. This is exactly what I typed:

> ```
> Design a spur gear pair for me and build it in FreeCAD. An electric motor
> delivers 3 kW at 1450 rpm into the pinion, I need a 3:1 reduction to the
> output, it runs most of the day driving a conveyor with light shock, and I
> would like it in a case hardened steel. Size it properly, check the teeth
> will not fail, and keep it reasonably compact.
> ```

The bridge was already connected to FreeCAD at this point. The helper could see its set of tools, so it was ready to both work out the design and build the geometry.



The bridge reporting as connected, with the five tools it uses to drive FreeCAD.

Step 2. How the helper thought about it

The first thing it did was sensible. Instead of guessing the missing details, it stopped and asked. It worked out everything it could from what I had given, then put the open decisions back to me as clear questions. The

important one was about how the small gear sits on the motor shaft, because that choice changes the whole gear. It even noticed that a bore and keyway of a certain size would leave too little solid metal under the teeth, and it offered two complete, self consistent designs rather than one.

```
Questions

1. Shafts: Does the pinion slide straight onto the motor shaft (Ø28 keyed), or onto its own input shaft or as a pinion cut
   into the shaft? What bore does the output gear need? I'd suggest Ø30 mm with a key for 58 N·m.
   - If it goes on the motor shaft, I'd use module 2 with 26 and 78 teeth: pinion 52 mm across, gear 156 mm, centre distance
     104 mm, face width 20 mm.
   - If it's a pinion shaft, it can stay at module 2 with 20 and 60 teeth: centre distance 80 mm.
2. Centre distance and ratio: Can I choose the centre distance, or does it have to fit an existing housing or mounting? Does
   the ratio need to be exactly 3.00?
3. Life and how it's made: Can I assume 16 h/day for 10 years (about 48,000 h, roughly 4×10⁸ pinion revolutions)? And can I
   assume the teeth are hobbed, case-hardened and then ground to ISO 1328 grade 6–7?

If you'd rather not decide, say "pick sensible values". I'll then use option 1 (on the motor shaft) with the defaults above
and report every value I chose.

FreeCAD 1.1.3 is connected, and a gear module (fcgear) imports. I'll make sure it really produces the tooth shapes before I
rely on it.

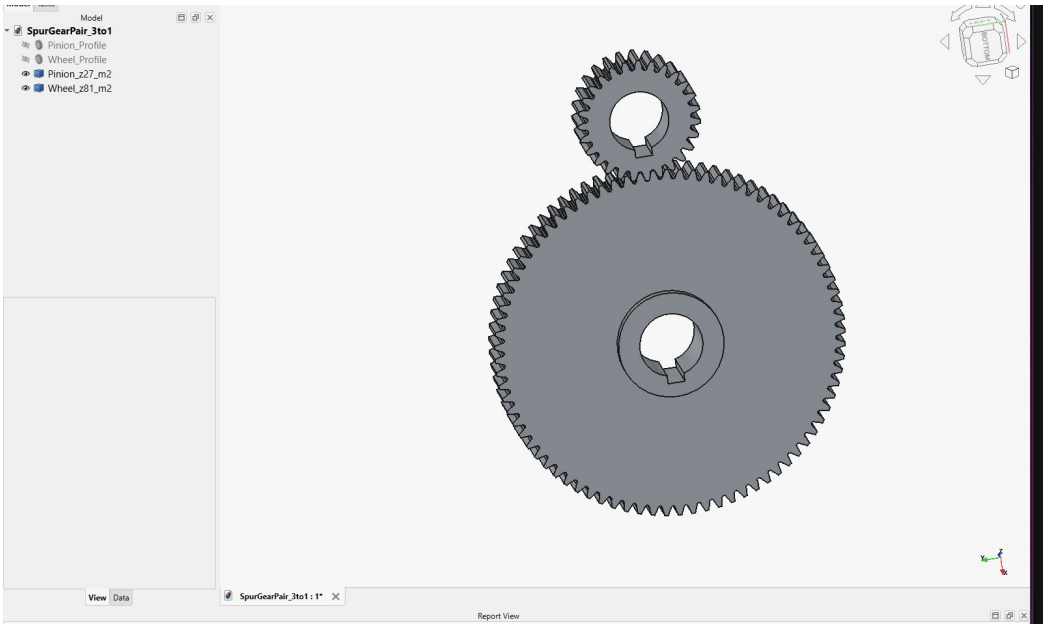
These figures are concept-level only. A gear going into production needs a certified ISO 6336 / AGMA rating, and the shafts,
bearings and housing need checking as well.
```

The helper gathering the few missing inputs before committing to any dimensions.

This is the behaviour I was hoping for. It treated the standard rules of gear design as a checklist, used torque and speed to pick a sensible tooth size, kept the pressure angle and tooth count within normal limits, and was honest that a full certified rating is a separate job. In short, it acted like a careful engineer gathering a brief, not like a tool that spits out the first thing that fits.

Step 3. The gear pair it built

Once the brief was clear, it modelled the pair in FreeCAD. It chose a tooth size of module 2, with 27 teeth on the small gear and 81 on the large one, which gives the exact three to one reduction asked for. Both gears came out as proper involute gears with real teeth, keyed bores for the shafts, and they mesh correctly at the right centre distance.



The finished pinion and wheel in FreeCAD, meshing at a 108 mm centre distance.

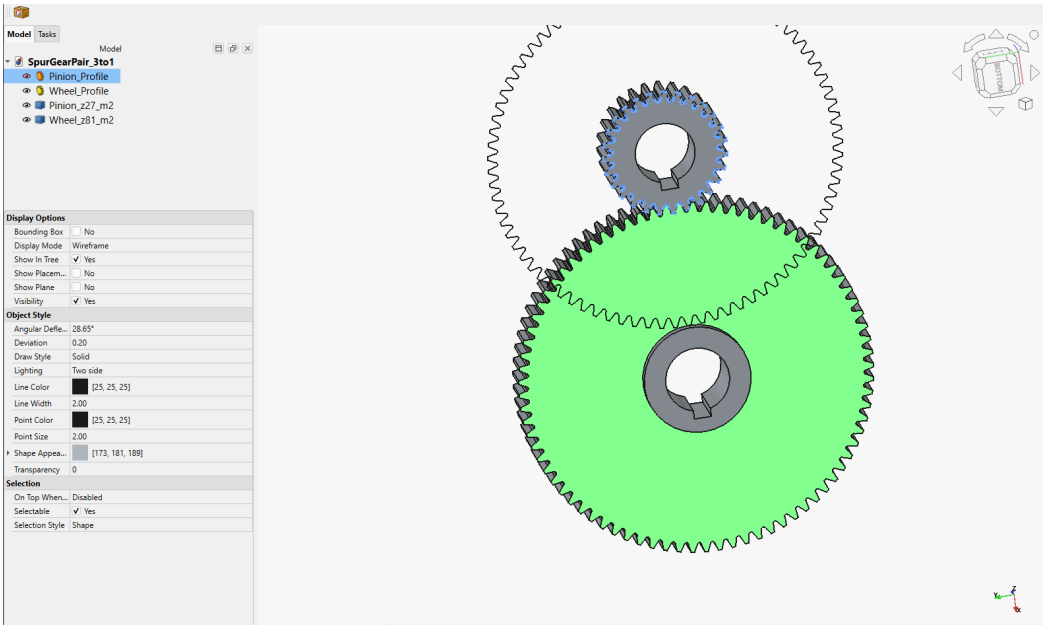
One detail is worth calling out. The small gear ended up far stronger than it needs to be for the load. That is not a mistake. The size of the shaft hole and its keyway set the smallest gear that could be used, so the gear is larger than the load alone would require. The helper said this plainly, which matters, because a large safety number can look wrong until you know why it is there.

The key numbers

Item	Value
Input power and speed	3 kW at 1450 rpm into the small gear
Reduction ratio	3.00 to 1 (exact)
Tooth size (module)	2 mm, 20 degree pressure angle
Tooth count	27 on the pinion, 81 on the wheel
Centre distance	108 mm
Face width	22 mm pinion, 20 mm wheel
Output speed and torque	483 rpm, about 58 Nm
Tooth force at the mesh	732 N
Material	Case hardened steel (16MnCr5 class)

It can be changed by hand later

A nice surprise was that the gears were not built as frozen shapes. They were made as live, editable objects. If I double click a gear in FreeCAD, a panel opens where I can change the number of teeth, the module, or the pressure angle, and the teeth are rebuilt. So I can adjust the design without going back to the AI every time. This answered an earlier worry I had about not being able to tweak a dimension once it was made.



Each gear keeps its design values editable in FreeCAD. The pinion shows 27 teeth, module 2, 20 degree pressure angle.

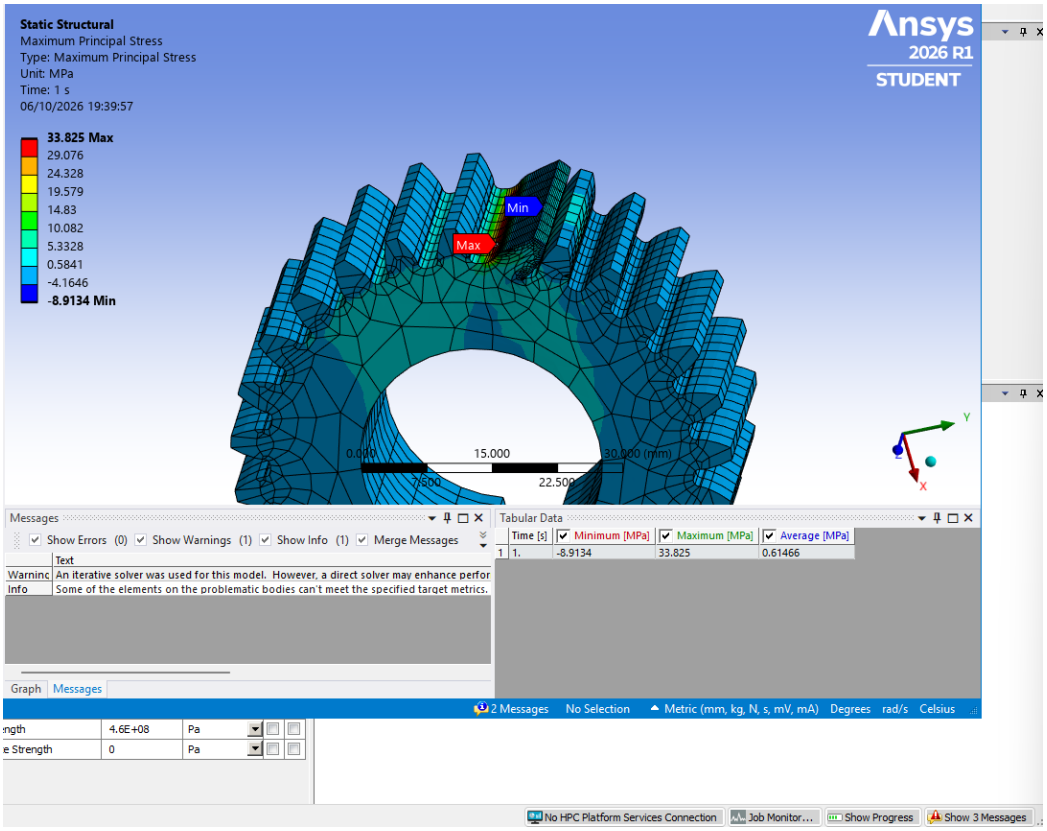
Step 4. Checking the result for real

A gear that looks right is not the same as a gear that is right. The helper gave a strength estimate using a simplified version of the standard gear method, and it was honest that some of the factors in it were read off charts rather than fully calculated. So I did not take its word for the strength. I exported the small gear and ran my own static structural analysis in ANSYS, which is independent of the helper and is the proper way to check tooth bending stress.

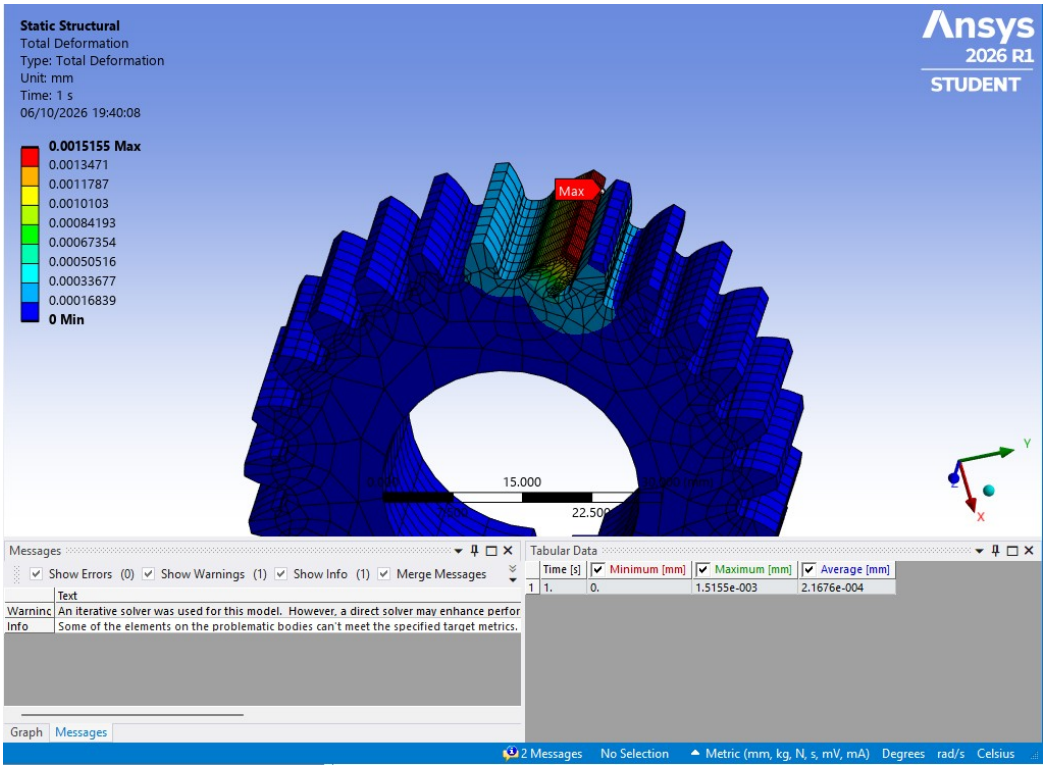
The test was set up the standard way. I held the gear at its shaft bore, put the full tooth force of 732 N on a single tooth near the tip, which is the worst case for bending, and refined the mesh at the base of that tooth where the stress peaks. The material stiffness used was that of steel.

What the simulation showed

The peak stress at the root of the loaded tooth came out at about 33.8 MPa, and the tooth tip moved by less than two thousandths of a millimetre. The stress sits exactly where tooth bending stress should sit, right at the base of the loaded tooth, and it dies away everywhere else.



ANSYS tooth bending result. Peak stress 33.8 MPa at the root of the loaded tooth.



Tooth deflection under the same load, about 0.0015 mm at the tip. Tiny, as expected for an over strong tooth.

Does it agree with the helper

Yes, and comfortably. The helper had estimated the clean tooth bending stress at around 52 MPa, before the extra safety factors that the standard method adds, and about 96 MPa with those factors in. My simulation gave about 34 MPa. These are all in the same range, and the simulation sitting a little lower is exactly what you expect, because the hand method is deliberately cautious and because my tip load direction was a simple, conservative choice.

Source	Tooth root stress	Meaning
Helper, before safety factors	about 52 MPa	Hand estimate
Helper, with safety factors	about 96 MPa	Cautious design figure
My ANSYS simulation	about 34 MPa	Independent check
Material limit (16MnCr5)	about 430 MPa	What the steel can take

Every one of those stress numbers is far below the 430 MPa the steel can handle. So the conclusion that the gear is strong, with a very large margin, is confirmed from two directions. The large margin is real and expected, because, as noted, the shaft hole set the gear size, not the load.

What was checked and what was not

To keep this honest, here is the line between what I proved and what I did not.

Checked:

- The gear geometry is correct. FreeCAD confirms the two gears are valid solids that mesh with no overlap, and the tooth count, module and pressure angle match the design.

- The tooth bending strength is sound. My own simulation matches the helper's estimate and shows a large margin against the material limit.

Not checked:

- Surface wear of the teeth over many years, and fatigue life, which need different methods and test data.
- The shafts, bearings and housing that the gears sit in.
- A formal certified rating to the full gear standard, which is a separate and larger piece of work.

Conclusion

The helper took a short, slightly open brief and turned it into a correct, editable, sensibly sized gear pair. It asked the right questions first, it was honest about which of its numbers were estimates, and when I checked the most important of those numbers with my own simulation, it held up with room to spare.

For this kind of job, the helper works well. It is a fast and careful first pass that gets you to a real, checkable design, and a human still closes the last step with proper simulation and, for a product, a certified calculation. That split is the right one, and this gear study is a clean example of it working end to end.